

1 Power series

Question 1. Determine the interval of convergence of the following power series.

1. $\sum_{n=0}^{\infty} \frac{x^n}{3^n}$

2. $\sum_{n=0}^{\infty} \frac{(x-1)^n}{3^n}$

3. $\sum_{n=0}^{\infty} \frac{x^{2n}}{3^n}$

4. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

5. $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$

6. $\sum_{n=1}^{\infty} \frac{x^n}{2 \cdot 4 \cdot 6 \cdots (2n)}$

7. $\sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{n} x^n$

8. $\sum_{n=0}^{\infty} \frac{x^n}{\sqrt{2n+1}}$

2 Maclaurin/Taylor series

Question 2. Find the Maclaurin series (Taylor series centered at $x = 0$) for the following functions.

1. $f(x) = \frac{1}{3+x}$

2. $f(x) = e^{-x^2}$

3. $f(x) = (x+1)^3 + (x+1)^2 + (x+1) + 1$

4. $f(x) = \frac{e^x + e^{-x}}{2}$

5. $f(x) = \frac{d^2}{dx^2} (x \cos(x^2))$

6. $f(x) = \int_0^x \frac{\sin t}{t} dt$

7. (challenge) $f(x) = \int_{-2}^x \frac{1}{t^2+4} dt$